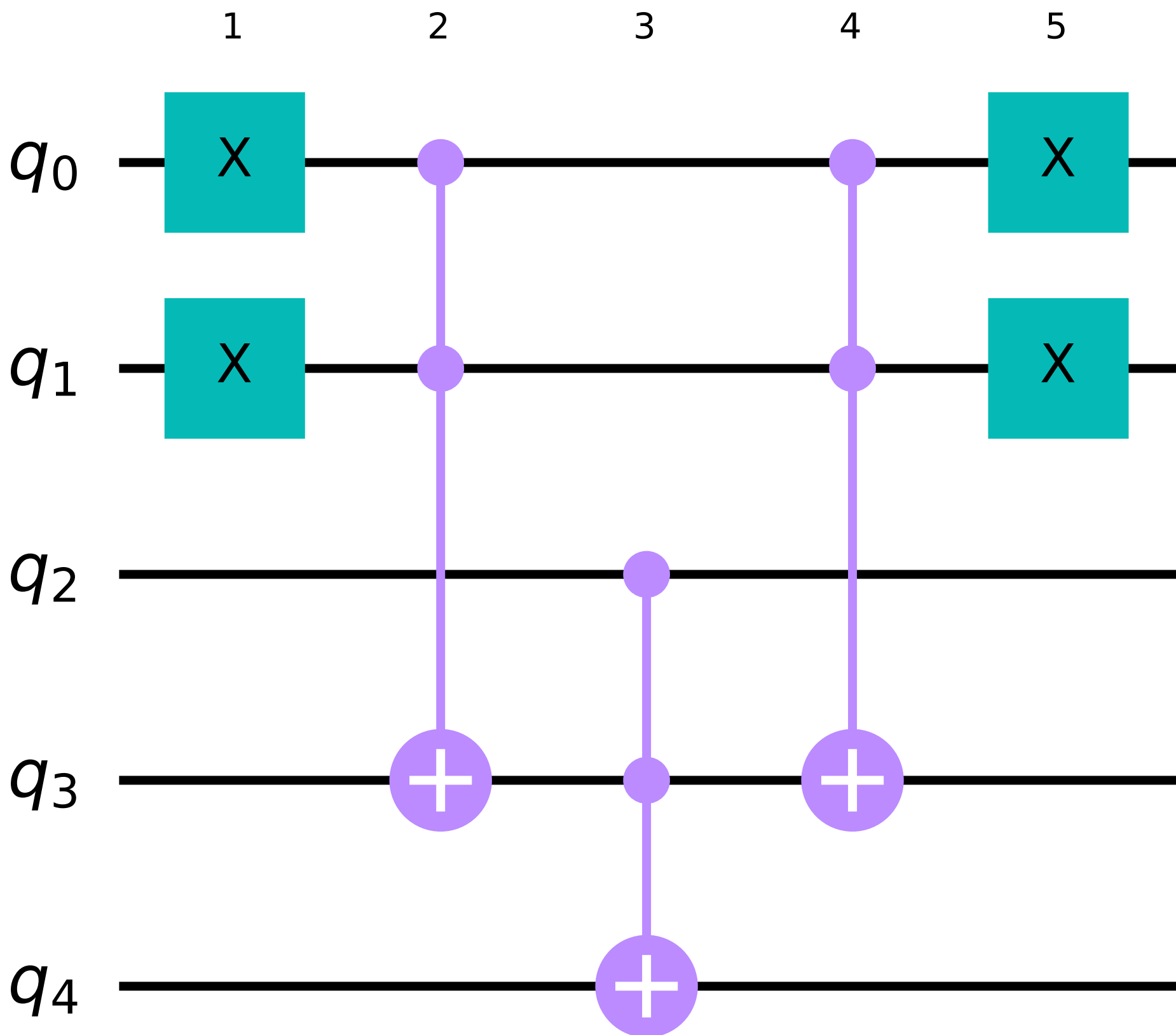


Problem 7: $S_3(x)$ Reversible Circuit (Toffoli + NOT)
via TensorCircuit → Qiskit → .draw()



$S_3(x) = \neg x_1 \wedge \neg x_2 \wedge x_3$
Gates: 4 NOT + 3 Toffoli = 7
Qubits: 3 input + 1 ancilla + 1 result = 5 Product: 7 × 5 = 35
Pipeline: TensorCircuit build_s3 → .to_qiskit() → qiskit.draw(output='mpl')